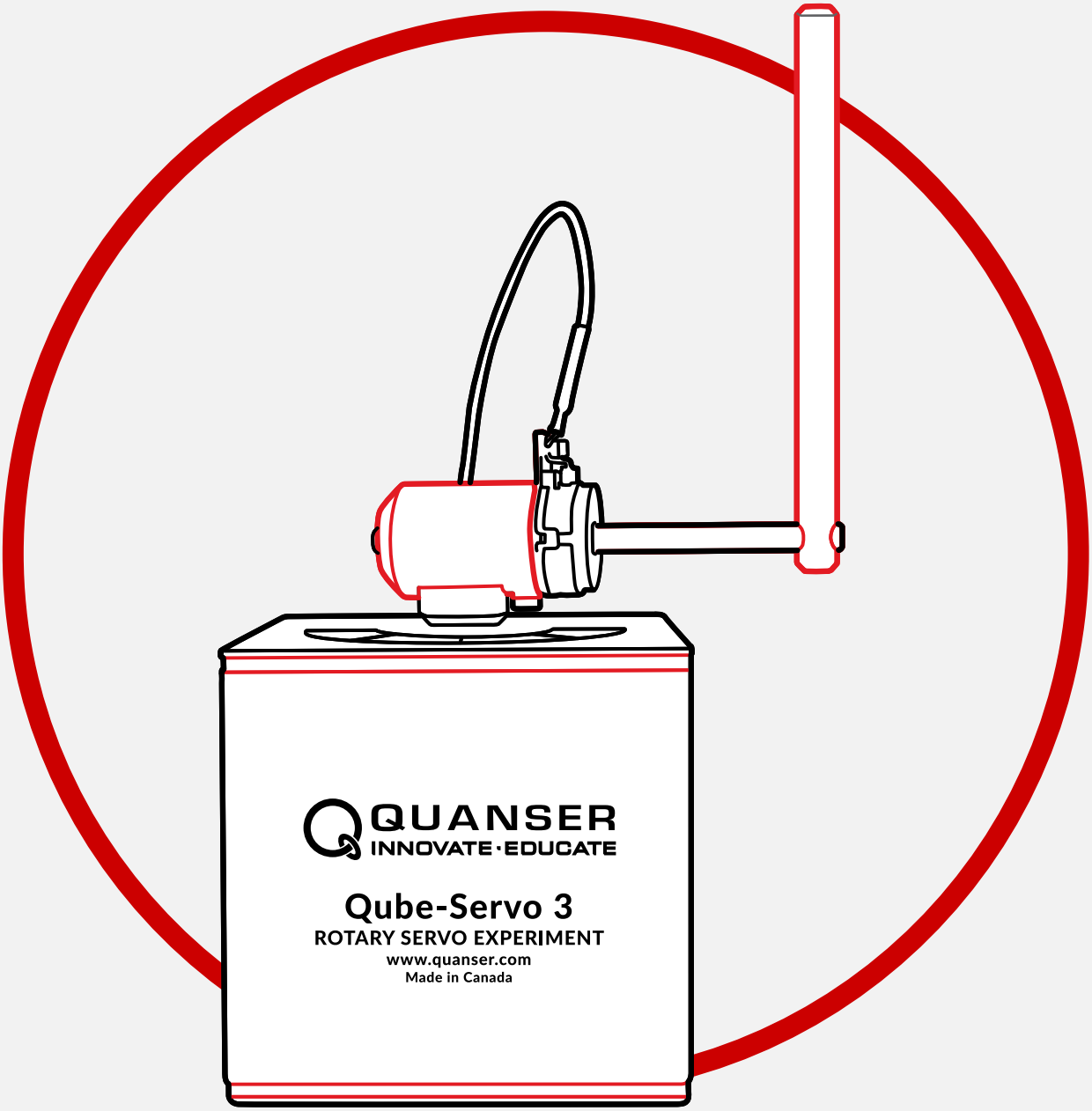

Qube-Servo 3

Block Diagram Modeling

V1.2 – 27th February 2025

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Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

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- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Qube-Servo 3 – Application Guide

Block Diagram Modeling

Why explore Block Diagram Modeling?

Block diagram modeling provides a visual way to represent how different parts of a DC motor system interact. Think of it as creating a flow chart showing how electrical input becomes mechanical output.

The diagram typically includes blocks for the electrical system (representing current response), mechanical system (representing motion), and various feedback paths like back-EMF. This modeling approach helps visualize the relationships between system components and is particularly valuable for understanding system behavior and designing control systems. In the case of a DC motor using these parameters and equations, the angular velocity can be predicted from the input voltage using a block diagram.

Background

This is one of multiple labs describing how to model a servomotor. Any of these labs can be done in any order. These modeling labs include modeling through frequency response, step response, parameter estimation, block diagrams and state space.

Prior to starting this lab, please review the following concept reviews (should be located in Documents/Quanser/4_concept_reviews/),

- Concept Review – Modeling → Modeling (Fundamental DC Motor Concepts section). You will need to understand the motor parameters as well as the mechanical and electrical equations.

Getting started

In this lab you will use the parameters found in the user manual and DC motor equations of motion from the concept review to create a block diagram model of the system response of the Qube-Servo 3.

Ensure you have completed the following labs

- **Hardware Interfacing Lab**
- **Filtering Lab**

Before you begin this lab, ensure that the following criteria are met.

- If using a physical Qube-Servo 3, make sure it has been setup and tested. See the Qube-Servo 3 Quick Start Guide for details on this step. Make sure the inertia disc load is attached to the Qube-Servo 3.
- If using the virtual Qube-Servo 3, make sure you have Quanser Interactive Labs open in the Qube 3 - DC Motor → Servo Workspace.
- **You have the Qube-Servo 3 User Manual.** It will be required to find relevant system information for some of the exercises.
- You are familiar with the basics of Simulink. See the Simulink Onramp for more help with getting started with Simulink.